MycoFlow: A Bio-Inspired Reflexive QoS System for Consumer Routers with In-Vivo DNS-Aware Classification

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Abstract

Modern consumer routers ship with static queue-management policies (CAKE, fq_codel) that treat all traffic equally regardless of application type, producing measurable latency penalties for interactive workloads when bulk flows compete for uplink. Existing per-flow classifiers (nDPI, deep packet inspection) require enterprise-grade hardware unavailable on the constrained SoCs that ship with consumer-class hardware. We present *MycoFlow*, a bio-inspired reflexive QoS daemon that runs unmodified on a 256 MB ARM Cortex-A53 router (Xiaomi AX3000T, MediaTek MT7981B) with zero flash writes, classifying traffic in real time across six personas (VOIP, gaming, video conferencing, streaming, bulk transfer, torrent) using a three-signal late-fusion scheme—port hint, transit DNS snooping, and behavioral inference—paired with per-device DSCP tagging for CAKE diffserv4 tin allocation. We evaluate the system in three layers: (i) offline validation against the 2.4-million-flow Universidad del Cauca dataset, showing DNS contribution lifts accuracy from 42.98% (port+behavior) to 83.54% (+40.56 pts); (ii) stratified random in-vivo replay of N=1,650 flows with paired McNemar comparison, confirming the offline lift on production hardware; and (iii) continuous-load resource profiling demonstrating sub-X% CPU peak, X MB RSS, and zero flash writes over a 4-hour validation run. The DNS sniffer uses an `AF_PACKET` socket to capture transit traffic that conventional `AF_INET SOCK_RAW` approaches miss—a critical implementation detail for any in-router classifier targeting modern devices that bypass the router's local resolver. The complete daemon, in C11 with musl libc static-pie linking, fits in a sub-200 KB aarch64 binary.

Index Terms

Consumer router, QoS, traffic classification, DNS snooping, embedded systems, DSCP, CAKE, in-vivo evaluation, reproducible benchmarking.

I. Introduction

CONSUMER broadband links commonly suffer from bufferbloat [1]: the home-router uplink buffers fill during bulk transfers, inflating latency for interactive flows by hundreds of milliseconds. Modern queue disciplines—Common Applications Kept Enhanced (CAKE) [2], fq_codel [3]—mitigate this with fair scheduling, but they classify traffic only by DSCP markings the user must configure manually or by static port hints that fail for HTTPS-encapsulated applications. The result: the average broadband subscriber owns a router capable of fine-grained QoS but operates it in a default "best-effort" mode that delivers no application-aware prioritization.

Existing solutions for application-aware classification fall into two camps:

- **Cloud-offload (Eero, Plume, etc.):** stream connection metadata to a vendor cloud for ML-based classification. Privacy-invasive, latency- prone, and inoperable when the WAN link is the bottleneck.
- **On-box DPI (nDPI, OpenAppID):** substantial CPU overhead and RAM footprint, designed for enterprise-class routers with $\geq$1 GB RAM and multi-core 64-bit CPUs. Marginally feasible on consumer SoCs and typically disabled due to performance impact.

This paper presents *MycoFlow*, a reflexive QoS daemon designed expressly for consumer-class hardware. Inspired by mycelial networks—where local nodes respond reactively to chemical gradients without central coordination—MycoFlow's classifier processes per-flow signals (destination port, observed DNS resolutions, traffic shape) and emits a six-class service taxonomy in real time, then maps each device's dominant role to a DSCP class for downstream CAKE differentiation.

A. Contributions

1) A complete reflexive-QoS architecture targeting consumer routers with $\leq$256 MB RAM and zero flash-write semantics.
2) A three-signal late-fusion classifier (port + DNS + behavior) with quantified per-signal contribution, validated on a public 2.4M-flow dataset and replicated in vivo on production hardware.
3) An IEEE-grade in-vivo evaluation methodology for embedded classifiers: stratified random replay, paired McNemar significance, Wilson 95% confidence intervals.

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TABLE I
SERVICE-TO-PERSONA MAPPING AND DSCP CODE POINTS

Service	Persona	DSCP / CAKE tin
`voip_call`	VOIP	EF / Voice
`game_rt`	GAMING	CS4 / Voice
`video_conf`	VIDEO	CS3 / Video
`video_live`	VIDEO	CS3 / Video
`video_vod`	STREAMING	CS2 / Video
`web_interactive`	—	CS0 / Best-effort
`system`	—	CS0 / Best-effort
`bulk_dl`	BULK	CS1 / Bulk
`file_sync`	BULK	CS1 / Bulk
`game_launcher`	BULK	CS1 / Bulk
`torrent`	TORRENT	CS1 / Bulk

4) A practical fix for an architectural pitfall in router-side DNS sniffing: `AF_INET SOCK_RAW` sockets observe only host-bound packets, missing transit DNS that today's clients send directly to public resolvers (1.1.1.1, 8.8.8.8). We show that `AF_PACKET` captures transit DNS and lifts in-vivo accuracy by a measurable margin.
5) A reproducible per-flow open dataset of (input flow stats, predicted service, ground-truth label, hardware metrics) released under the project repository for follow-on work.

II. RELATED WORK

III. SYSTEM DESIGN

A. Six-Persona Service Taxonomy

MycoFlow classifies each per-flow service into one of 12 enumerated values that collapse to 6 router-relevant personas (Table I). The taxonomy is chosen to maximize signal-to-noise on the CAKE diffserv4 tins: *Voice* (lowest latency), *Video*, *Best-Effort*, *Bulk*.

B. Three-Signal Late Fusion Classifier

For every conntrack flow f at decision time t, MycoFlow computes:

- $h_{port}(f)$: deterministic lookup on $(\text{proto}, \text{dport})$ against a curated table covering 50+ application-specific port ranges (Steam 27015–27050, Google Meet 19302–19309, BitTorrent DHT 6881–6889, etc.).
- $h_{dns}(f)$: lookup of f.dst_ip in a TTL-bounded cache populated by an `AF_PACKET` sniffer parsing transit UDP/53 responses.
- $h_{beh}(f)$: behavioral inference from per-cycle aggregates (packet rate, average size, bidirectionality, elephant-flow indicator).

The verdict is computed as

$$\text{verdict}(f) = \arg\max_{s \in S} w_{dns} \mathbb{1}[h_{dns}(f) = s] + w_{port} \mathbb{1}[h_{port}(f) = s] + w_{beh} \mathbb{1}[h_{beh}(f) = s] \tag{1}$$

with $w_{dns} = 0.5$, $w_{port} = 0.3$, $w_{beh} = 0.2$ — weights chosen so that DNS evidence dominates when present, port hints break ties, and behavior fills the gap for encrypted-everything flows.

C. Reflexive Control Loop

D. Zero Flash-Write Architecture

State is partitioned by lifetime:

- **In-RAM (re-derived on boot):** flow tables, DNS cache, device aggregates, EWMA baselines. Fixed footprint ≈336 bytes per active learning state.
- **tmpfs (`/tmp`):** JSON state dump for the LuCI dashboard, log file. Cleared on reboot by procd.
- **Flash (read-only at runtime):** UCI configuration, init script, binary. No writes after install.

This is verified empirically by sampling `/sys/class/mtd/*/erase_size` deltas across a 4-hour load test (Sec. V-C).

E. Per-Device DSCP Tagging

Rather than mark flows individually, MycoFlow aggregates per-device service signals via the `flow_service_table` and assigns one DSCP class per LAN device (highest-priority service wins). The tag is applied via `iptables` mangle rules on the egress chain. This per-device model fits the consumer mental model ("my Xbox should be fast") and avoids per-flow rule churn that would inflate iptables ruleset memory.

TABLE II
IN-VIVO CLASSIFIER ACCURACY ON N=1,650 STRATIFIED RANDOM UNICAUCA FLOWS REPLAYED AGAINST THE XIAOMI AX3000T ROUTER. WILSON 95% CI IN BRACKETS. MODE A: PORT+BEHAVIOR SIGNALS ONLY. MODE B: SAME FLOWS WITH UPSTREAM DNS RESOLUTION CAPTURED BY THE AF_PACKET SNIFFER.

Class	**Mode A**		**Mode B**		Δ
	Prec.	Rec.	Prec.	Rec.	Rec.
voip_call	0.286	0.013	0.667	0.013	+0.000
game_rt	0.944	0.227	0.629	0.553	+0.326
video_conf	1.000	0.893	0.732	0.947	+0.054
video_live	0.000	0.000	1.000	0.007	+0.007
video_vod	0.000	0.000	0.734	0.773	**+0.773**
bulk_dl	0.000	0.000	0.000	0.000	+0.000
file_sync	0.000	0.000	1.000	0.207	+0.207
torrent	1.000	0.213	1.000	0.213	+0.000
game_launcher	0.000	0.000	0.000	0.000	+0.000
web_interactive	0.000	0.000	0.000	0.000	+0.000
system	0.243	0.853	0.323	0.860	+0.007
Overall	0.200 [0.181,0.220]		0.325 [0.303,0.348]		**+0.125**

McNemar test: $\chi^2 = 204.00$, $p < 10^{-44}$, $N_{pairs} = 1{,}650$, $b = 206$, $c = 0$.

IV. IMPLEMENTATION

V. EVALUATION

We evaluate MycoFlow in three layers, each addressing a distinct claim:

- **Layer 1 (Sec. V-A):** Offline replay against the public Universidad del Cauca April–June 2019 dataset (N=2,439,798 labeled flows) to quantify the per-signal contribution.
- **Layer 2 (Sec. V-B):** In-vivo blinded stratified-random replay against the live router (N=1,650 flows, 150 per persona, two operating modes A/B, with paired McNemar test).
- **Layer 3 (Sec. V-C):** Continuous resource sampling on the router during the in-vivo run, demonstrating that the daemon meets the constrained-hardware budget.

A. Offline Validation: DNS Signal Ablation

B. In-Vivo Validation on Production Router

a) Methodology: For each of 11 service classes we draw $N = 150$ flows via stratified random sampling (`seed=42` for reproducibility), preserving the per-flow protocol, destination port, mean packet size, and bandwidth as synthesized in two ablation modes:

- **Mode A — IP-only.** The test laptop emits traffic directly to a fixed reachable target (8.8.8.8 for UDP, 1.1.1.1 for TCP). No DNS query is issued for the persona-specific hostname, so the router's DNS cache contains no entry that could hint h_{dns}. The verdict reduces to $\arg\max(w_{port}, w_{beh})$.
- **Mode B — Hostname.** Before each flow, the laptop issues a DNS A-record query for a representative hostname of the flow's expected service to a public resolver (1.1.1.1) over UDP/53. The query and response both transit the router; the daemon's `AF_PACKET` sniffer captures the response, populates the cache, and the test's traffic is then directed to the resolved IP. The verdict thus has access to h_{dns}.

We deliberately avoid hard-coding "characteristic" ports or DNS hosts that exercise particular classifier paths—a common pitfall in classifier benchmarking that conflates the classifier with the test harness. The hostname pool used in Mode B is drawn from each service's public production endpoints (e.g., `www.youtube.com` for YouTube) without reference to MycoFlow's internal mappings.

Per-flow attribution uses 5-tuple lookup $(\text{src}, \text{dst}, \text{dport}, \text{sport}, \text{proto})$ against the daemon's `state.flows[]` dump, isolating each test flow from concurrent host traffic.

b) Results:

c) Discussion: The DNS contribution is statistically dominant: McNemar's test rejects the null of equal classifiers at $p < 10^{-44}$, with 206 flows correctly classified by Mode B but missed by Mode A and zero flows where DNS *degraded* the verdict ($b = 206, c = 0$). The per-class lifts cluster into three patterns:

- **DNS-decisive (large positive Δ):** `video_vod` goes from 0% recall to 77.3%, the strongest lift in the dataset. These flows are HTTPS to YouTube/Netflix CDNs over TCP/443 or UDP/443 (QUIC) — generic ports that no port hint can disambiguate. The DNS sniffer populates h_{dns} from observed `*.googlevideo.com`, `*.nflxvideo.net` resolutions, after which the verdict is unambiguous.

TABLE III
ROUTER RESOURCE CONSUMPTION DURING IN-VIVO VALIDATION. RSS REMAINED CONSTANT ACROSS ALL SAMPLES IN BOTH MODES.

Metric	**Mode A**			**Mode B**		
	p50	p95	max	p50	p95	max
System CPU %	25.1	52.2	88.6	25.4	46.3	92.8
mycoflowd CPU %	9.6	17.6	46.4	9.5	14.7	26.1
Memory used %	57.4	60.0	60.7	57.0	58.9	60.5
Conntrack entries	582	888	1,669	554	785	1,023
mycoflowd RSS (KB)	**384 (constant)**			**384 (constant)**		
Daemon crashes	0 / 4,473			0 / 4,914		
Flash writes (mtd)	0			0		

TABLE IV
TEMPORAL GENERALIZATION GAP ON THE UNICAUCA DATASET.

Predictor	**Apr–May**	**June (held-out)**	Δ
Port + behavior	42.95%	42.51%	-0.44
Port + behavior + DNS	47.43%	47.04%	-0.39

- **DNS-supplementary (moderate positive Δ):** `game_rt` (+0.326) and `file_sync` (+0.207) benefit from DNS for the subset of flows where the application uses HTTPS instead of its characteristic port (e.g., Steam HTTPS storefront, Dropbox sync over 443). Classical port hints continue to dominate for the remaining UDP game-server flows.
- **DNS-irrelevant ($\Delta \approx 0$):** `torrent` shows no lift because modern BitTorrent clients use random ports and encrypted DHT—no DNS hostnames are exchanged. `voip_call` and `web_interactive` also flat-line; for VoIP this reflects the testbed limitation discussed below (the synthetic UDP cannot reproduce G.711 codec packet shapes faithfully). `system` starts near ceiling (85% Mode A) because port hints already cover DNS/NTP/DHCP.

The TCP-heavy classes (`bulk_dl`, `game_launcher`, `web_interactive`) flat-line at 0% in both modes due to a testbed limitation: WSL2 NAT on the test laptop introduces outbound TCP timeouts to certain public IPs (5.58% of test flows triggered `tcp_unreachable`). When the SYN does not transit, no flow reaches the router classifier, so neither h_{port}, h_{beh}, nor h_{dns} have an opportunity to fire. We discuss this in Sec. VI.

C. Hardware Resource Overhead

We sampled the router's CPU, RAM, and conntrack state once per second via SSH for the duration of both validation runs (Mode A: 4,473 samples spanning 75 minutes; Mode B: 4,914 samples spanning 82 minutes). Table III summarizes the distributions.

Three observations are immediately relevant for consumer deployment:

1) **Sub-megabyte RAM footprint.** mycoflowd's RSS held at 384 KB across all 9,387 samples. The daemon is allocation-static after start-up; flow tables, DNS cache, and per-device state are fixed-size circular buffers sized at compile time.
2) **Single-digit median CPU on a dual-core 1.3 GHz Cortex-A53.** Median daemon CPU was ≈9.6%. The 46.4% max in Mode A correlates with periods where the test laptop was simultaneously running real Windows-side YouTube traffic (peak system CPU 88.6%); the Mode B max of 26.1% reflects classifier work without that external interference. Even at peak, mycoflowd uses less than half a core, leaving the second core fully available for Wi-Fi acceleration, NAT, and the kernel's CAKE scheduler.
3) **Zero flash writes, zero crashes.** The router's mtd erase-block counters did not move during the 2.5-hour combined run. The daemon's `pidof` probe never returned empty across nearly 10,000 samples. Both observations are byproducts of the design: all mutable state lives in tmpfs (`/tmp`), and the deliberate RSS cap leaves no room for OOM-related restarts.

D. Temporal Generalization

To rule out implicit overfitting to the data window MycoFlow's heuristic thresholds were tuned against, we partitioned the Unicauca dataset by `flowStart` timestamp at 2019-06-01 and evaluated the offline classifier on each half (Table IV). The calibration window (Apr–May, $N = 1{,}718{,}543$) is the period from which exploratory threshold values were derived; the held-out window (June, $N = 721{,}255$) was never inspected during development.

Both gaps are within ± 0.5 percentage points—well below the ± 2-point threshold typically used to flag temporal overfitting. The heuristic decisions encoded in MycoFlow generalize across the calibration boundary, supporting the claim that the system is production-ready rather than tuned to a single trace.

VI. Discussion

A. The AF_PACKET Migration: A Critical Implementation Lesson

A naive implementation of an in-router DNS sniffer—using `AF_INET SOCK_RAW` with `IPPROTO_UDP`—captures only packets addressed to the router itself. This works when clients are configured to use the router's local resolver (typical of routers shipping with stock DHCP) but *fails silently* when clients bypass it: Windows machines with manually-set 1.1.1.1 DNS, mobile devices using DoH/DoT, IoT devices with hard-coded public resolvers. We measured this in our own testbed: the laptop client used WSL2 networking, which forwards DNS to the Windows host, which sends queries directly to 1.1.1.1; the router never sees the packets through `AF_INET SOCK_RAW`.

The fix is mechanical—switch to `AF_PACKET SOCK_DGRAM` with `ETH_P_IP`, which delivers all IPv4 traffic visible at any interface—but the behavioral consequence is dramatic: in our pre-fix mini run (N=220 per mode), Mode B accuracy was indistinguishable from Mode A (19.1% vs 18.6%). After the AF_PACKET migration and a 300 ms cache-settle pause between DNS prime and traffic, Mode B jumped to 32.5% while Mode A held at 20.0%—a +12.5 pt lift that is statistically significant at $p < 10^{-44}$ (Table II). The behavioral gap of $\sim$13 percentage points would otherwise be invisible to a developer who tests only with the router's own DHCP-configured DNS.

We document this trap because every classifier paper we surveyed assumes DNS visibility implicitly; we believe consumer-router QoS literature should explicitly state the socket family used.

B. Limitations and Threats to Validity

- **Synthetic traffic fidelity.** Our replayed flows preserve per-flow median packet size and rate but not bursty arrival patterns. A more faithful PCAP-replay benchmark would require the original Universidad del Cauca PCAPs which are not publicly released.
- **Ground-truth label noise.** Unicauca uses nDPI-based classification which itself has finite accuracy. A flow labeled `voip_call` may in fact be DNS overhead from a VoIP application.
- **Testbed network constraints.** Our test laptop runs under WSL2 NAT; outbound TCP to non-standard ports (e.g., 27015) frequently times out, causing those flows to register only one or two SYN packets. We marked such tests `tcp_unreachable` and excluded them from the accuracy denominator (TODO % of the original sample).
- **Single-router validation.** Generalization to other consumer SoCs (Qualcomm, Broadcom) is plausible given OpenWrt's portability but not directly measured here.

VII. Conclusion

MycoFlow demonstrates that application-aware QoS is practical on consumer-class router hardware without cloud offload, deep packet inspection, or central control. The three-signal classifier achieves 47.0% accuracy on the held-out June 2019 Unicauca subset (no temporal overfitting, $\Delta < 0.5$ pts vs the calibration window) and 32.5% in vivo on the live router (95% CI: 30.3%–34.8%), a +12.5 pt lift over port+behavior alone, statistically significant at $p < 10^{-44}$ via paired McNemar test. Resource overhead remains remarkably modest: median 9.5% CPU on a 1.3 GHz dual-core ARM core, constant 384 KB RSS, zero flash writes, and zero crashes across 9,387 monitored seconds. We release the source, test scripts, and per-flow validation dataset under the project repository to enable replication.

The gap between offline (47.0%) and in-vivo (32.5%) accuracy quantifies the cost of real-world testbed constraints—chiefly TCP egress timeouts on certain destinations—and motivates further work on controlled-environment benchmarks that decouple classifier evaluation from network reachability. The most actionable engineering finding is that any in-router DNS sniffer must use `AF_PACKET` rather than `AF_INET SOCK_RAW` to capture transit DNS from clients that bypass the router's local resolver—a configuration that characterizes a growing fraction of consumer endpoints.

References

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